

Midterm - CS393

1) $H \otimes I \cdot I \otimes Y \cdot CNOT(1, 0)$

$\rightarrow |0\rangle \otimes |0\rangle$ $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \cdot |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

$H \otimes I$

$H \otimes I$

$$\begin{cases} Y|0\rangle = i|1\rangle \\ Y|1\rangle = -i|0\rangle \end{cases}$$

$I \otimes Y \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \otimes Y|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes i|1\rangle + |1\rangle \otimes -i|1\rangle)$

$= \frac{1}{\sqrt{2}}(i|01\rangle - i|11\rangle)$

$(|0\rangle \otimes |1\rangle) \oplus (|0\rangle \otimes -|1\rangle)$

$\propto CNOT \Rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) =$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$

$I \otimes Y = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{bmatrix}$

$CNOT(\alpha, \beta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} H \cdot (Y \cdot Cx)$

$Y \cdot Cx = \begin{bmatrix} 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & 0 & 0 & i \end{bmatrix} \cdot H = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}$

6) A matrix is unitary if multiplied with its adjoint = identity

$U^* = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$\boxed{2} H \otimes I \cdot I \otimes Y \cdot CX(1,0) = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \quad \boxed{3} P(|110\rangle) = 0.0$$

Probability for $|00\rangle$: 0

$$\text{Final State: } \frac{1}{\sqrt{2}}(|0\rangle|01\rangle + |1\rangle|11\rangle)$$

Probability for each is Magnitude² of their Coeff.

$$\boxed{4} H \otimes I \cdot I \otimes Y \otimes I \cdot CX(1,0) \cdot CCX(0,1:2)$$

$$\parallel \begin{matrix} |0\rangle \otimes |0\rangle \otimes |0\rangle \\ \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |00\rangle \end{matrix} = \frac{1}{\sqrt{2}}(|000\rangle + |100\rangle)$$

$$I \otimes Y \otimes I \cdot \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}(|010\rangle + |110\rangle)$$

$$CX(0,1) \cdot \frac{1}{\sqrt{2}}(|010\rangle + |100\rangle) \rightarrow \text{Matrix in .txt}$$

$CCX(0,1:2) \rightarrow$ Has no affect $\therefore$ Same probabilities, as $\boxed{2}, \boxed{3}$ w/ $|0\rangle$ tensored (appended)
 $\hookrightarrow$ for $|000\rangle$ input.

$$\boxed{5} \quad \boxed{6} \quad \begin{matrix} \searrow \\ P|000\rangle : 0.0 \quad P|111\rangle : 0.0 \end{matrix}$$

for full prob list, see .txt.

7 $S: \{0,1\}^3 \rightarrow \{0,1\}^3$

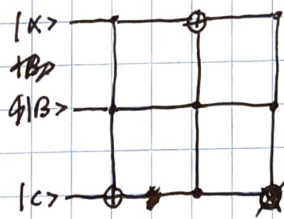
CCNOT(0,1,2)

$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$	
0	0	0	0	0
0	0	1	1	1
0	1	0	0	2
0	1	1	1	3
1	0	0	0	4
1	0	1	1	5
1	1	0	1	6
1	1	1	0	7

(a) Balanced.

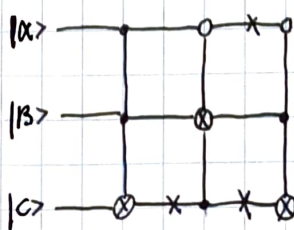
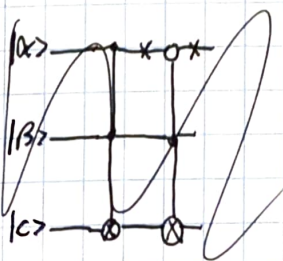


$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



b.1 - Neither

$ x\rangle$	$ y\rangle$	$ z\rangle$
0	1	0
1	0	1
2	1	0
3	0	0
4	0	0
5	0	1
6	0	0
7	1	1



$ x\rangle$	$ y\rangle$	$ z\rangle$	$ y\rangle$	
0	0	0	1	
0	0	1	1	
0	1	0	1	
0	1	1	1	
1	0	0	1	
1	0	1	1	
1	1	0	1	
1	1	1	1	

b.2 Constant

8 $S = 01 \quad \vec{x} = \vec{y} \text{ or } \vec{x} = f(x) \oplus \vec{s} \quad f(00) \mapsto 10 = f(01)$

a) How many f for $S = 01$? (Permutation)
 $f(00) = f(01) = 00$

$f(01) \mapsto 10$
 $f(10) \mapsto 00 = f(11)$
 $f(11) \mapsto 00$

4 possible outputs in $\{0, 1\}^2 \rightarrow \{0, 1\}^2$
 $00, 01, 10, 11$

$P(n, r)$ where n is number of choices, and r is how many containers.

$$P(n, r) = \frac{n!}{(n-r)!}$$

Containers: $f(00) = f(01)$ & $f(10) = f(11)$

$\therefore \# \text{ choices} = \frac{4!}{2!} = 12$

b) Space of vectors ortho to $\vec{s} \in \{0, 1\}$

~~$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0+1 = 1$~~

for $\mathbb{F}_2$, $1+1=0$ Orthogonal if dot product = 0

$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0+1=1 \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0$

$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \quad \therefore \perp: \{00, 10\}$

Also: $\{00, 01, 10, 11\} / \{00, 01\} \text{ mod } 2 = \{00, 10\}$
 Quotient Space

~~$\begin{matrix} 00 & 01 & 10 & 11 \\ 00 & 00 & 00 & 01 \\ 00 & & & \end{matrix}$~~

⑥

$$|00\rangle \otimes |00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \otimes \frac{1}{\sqrt{2}}|00\rangle$$

$$\frac{1}{2}(|00\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |01\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |10\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |11\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle))$$

$$\frac{1}{2}(|00\rangle \otimes |00\rangle + |01\rangle \otimes |00\rangle + |10\rangle \otimes |00\rangle + |11\rangle \otimes |00\rangle)$$

$$\begin{aligned} S(00) &\mapsto 10 \\ S(01) &\mapsto 10 \\ S(10) &\mapsto 00 \\ S(11) &\mapsto 00 \end{aligned}$$

$$\frac{1}{2}(|0010\rangle + |0110\rangle + |1000\rangle + |1100\rangle)$$

$H \otimes I$

$$\frac{1}{2}(H(|00\rangle)|10\rangle + H(|01\rangle)|10\rangle + H(|10\rangle)|00\rangle + H(|11\rangle)|00\rangle)$$

$$\frac{1}{2}(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle \otimes |0\rangle + |1\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle \otimes |0\rangle - |1\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle \otimes |0\rangle + |1\rangle)|00\rangle + \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle \otimes |0\rangle - |1\rangle)|00\rangle)$$

$$\frac{1}{\sqrt{2}}[|0\rangle - |1\rangle \otimes |0\rangle + |1\rangle]|00\rangle + \frac{1}{\sqrt{2}}[|0\rangle - |1\rangle \otimes |0\rangle - |1\rangle]|00\rangle$$

$$\frac{1}{2}[\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|10\rangle + \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)|10\rangle +$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)|00\rangle + \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)|00\rangle]$$

$$\frac{1}{2}[\frac{1}{\sqrt{2}}(|0010\rangle + |0110\rangle + |1010\rangle + |1110\rangle) \frac{1}{\sqrt{2}}(|0010\rangle - |0110\rangle + |1010\rangle - |1110\rangle) +$$

$$\frac{1}{\sqrt{2}}(|0000\rangle + |0100\rangle - |1000\rangle - |1100\rangle) + \frac{1}{\sqrt{2}}(|0000\rangle - |0100\rangle - |1000\rangle + |1100\rangle)]$$

$$\frac{1}{4}[2|0010\rangle - 2|0110\rangle + 2|1010\rangle - 2|1110\rangle + 2|0000\rangle - 2|0100\rangle - 2|1000\rangle - 2|1100\rangle]$$

$$\frac{1}{4}[|0000\rangle + |0100\rangle - |0100\rangle - |1000\rangle - |0110\rangle + |1010\rangle - |1100\rangle - |1110\rangle]$$

$$2|0010\rangle + 2|1010\rangle + 2|0000\rangle - 2|1000\rangle$$

$$\frac{1}{2}(|0010\rangle + |1010\rangle + |0000\rangle - |1000\rangle)$$

⑥ Given that:

$|0000\rangle \mapsto |0010\rangle$
 $|0100\rangle \mapsto |0110\rangle$
 $|1000\rangle \mapsto |1000\rangle$
 $|1100\rangle \mapsto |1100\rangle$

$$V_f =$$

0000 10> | 01 11> | 27 12> | 37 13> | 47 14> | 57 15> | 67 16> | 77 17> | 87 18> | 97 19> | 107 20> | 117 21> | 127 22> | 137 23> | 147 24> | 157 25> | 167 26>

0001 11> | 0010 27> | 0011 37> | 0100 147> | 0101 157> | 0110 167> | 0111 177> | 1000 187> | 1001 197> | 1010 1107> | 1011 1117> | 1100 1127> | 1101 1137> | 1110 1147> | 1111 1157> | 1200 1167>

8 1

After H:

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|00\rangle$$

$$\frac{1}{\sqrt{2}}: \frac{1}{2}(|00\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |01\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |10\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |11\rangle(|00\rangle + |01\rangle + |10\rangle + |11\rangle))$$

$$\frac{1}{2}(|00\rangle|00\rangle + |00\rangle|01\rangle + |00\rangle|10\rangle + |00\rangle|11\rangle + |01\rangle|00\rangle + |01\rangle|01\rangle + |01\rangle|10\rangle + |01\rangle|11\rangle + |10\rangle|00\rangle + |10\rangle|01\rangle + |10\rangle|10\rangle + |10\rangle|11\rangle + |11\rangle|00\rangle + |11\rangle|01\rangle + |11\rangle|10\rangle + |11\rangle|11\rangle)$$

H:

↓ extract 1st reg

$$\frac{1}{2}(|00\rangle + |01\rangle) = |00\rangle + |10\rangle$$

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \quad \frac{1}{2}(|01\rangle + |11\rangle) = |00\rangle - |10\rangle$$

Before Measurement, 1st reg = $\frac{1}{2}(|00\rangle + |10\rangle)$

② $\frac{1}{2}: |00\rangle, \frac{1}{2}: |10\rangle$